CST 300

Ethics Argument Essay

What is Hacking and why cybersecurity matter?

There are laws in place that allow you to exert a certain amount of force to defend yourself from harm. In the digital space, things are not so clear-cut. With the quick evolution of technology, the laws have not been able to create clear protocols or workflows to deal with cyber-attacks. To get a better understanding of what cyber-attacks consist of, it is important to define what hacking is. Hacking is a term that means to break into a digital device or system as an unauthorized user (Kosinski, 2025). Although breaking into another person's or organization's system may seem trivial, the possibilities are endless for a hacker. On one end of the spectrum, a hacker can access another person's or organization's system. On the extreme opposite end of the spectrum; data can be stolen, systems can be held ransom, or even it can even lead to the destruction of infrastructure. Because of this severe consequence, companies pay top dollar for cybersecurity specialists to protect their businesses, systems, and infrastructure. Being a victim of a cyberattack, the question then arises: If a person or organization should fall victim to a cyberattack, should that victim be given the right to defend themselves? This paper is meant to dive into the two stakeholders that are on opposite sides of this big question.

Stakeholder 1: The Cyber Attack Victim

The first stakeholder to be explored is the victim who fell victim to the cyber-attack. This stakeholder has either had their digital device or computer system(s) compromised. Cyber-attacks come in many shapes and sizes. For the sake of this argument, let us suppose that data was stolen. In today's digital landscape, stolen data means big money for hackers on the black market. Data can be used for identity theft, financial fraud, or even phishing attacks. In a journal

published by Google and the University of Berkeley, their teams measured how something as simple as a phishing tool can fool people into giving up their credentials. The journal states, "The most popular phishing kit—a file portal that supports "logins" from Yahoo, Hotmail, AOL, Gmail, and other mail providers—generated 1,448,890 reports of stolen credentials to 2,599 different exfiltration emails." (Thomas et al., 2017). Something as simple as credentials can lead to things such as data theft or even loss of access to one's system.

The stakeholder's position on the question is in support of the ability to hack back. It is in the best interest of the victim to support the ability to hack back. Stolen data can mean dire consequences for both an individual and an organization. The claims of the victim are built on moral reasoning: Something was taken without consent, and they are just retrieving what rightfully belongs to them. To hack back in the hopes of attribution, and getting the stolen data back, the next question is then, how much force is "too much" force? Drawing parallels to the physical world, the 2nd' amendment gives us the right to bear arms, and this resonates with claims of resemblance. Essentially, the idea of having the right to defend oneself should also be applied to the digital space. Conversely, it is not feasible to use lethal force in the hopes of attribution: in cyberspace, it is not morally correct to destroy the hacker's computer on the way out after retrieving lost data. The gray area thus begins as, what would be considered an acceptable level of "hacking back" to retrieve data and prevent it from leaking. These ideas will be explored further along with the stakeholder's argument.

Stakeholder 2: The Innocent Caught in the Crossfire

The second stakeholder is considered to be the third party caught in the crossfire of the cyber-attack. Often, hackers cleverly use cloaking tactics to hide their identity and location. According to an article from American University,

"Often, cyber threat actors will hijack unwitting victims' computers to carry out an attack. These computers could become collateral damage of a hack-back", (Winstead, 2020). Winstead highlights an issue that could be catastrophic to the world of cybersecurity legislation, innocent people and their systems getting damaged in the process of attribution and retaliation.

Consequently, the third parties involved in a cyber-attack are in opposition to hacking back. This stakeholder's main claim of policy is that if hacking back is legalized, it increases the risk of harming unintended innocent users. The possibility of an innocent party or parties getting their systems damaged without due process raises many concerns for the stakeholders. The secondary claim of cause revolves around the increased amount of collateral damage. If hacking back were to be legal, the third parties caught in the crossfire of a cyber-attack would face increased risk. Systems may be intentionally targeted or wrongfully exploited. Businesses and organizations that are seeking both attribution and relation could cause unintentional data breaches, operational disruptions, or even wrongfully destroying digital infrastructure.

The Big Question: Should the victim of a cyber-attack have the right to hack-back?

Stakeholders and their stances defined, the big question arises: Should victims of a cyber-attack have the right to hack-back? Hacking back in self-defense raises many concerns. Although self-defense makes a valid argument when under attack in the physical world, the same doesn't translate effectively in the digital space. In cyberspace, accountability is very difficult to pinpoint. Allowing parties to take matters into their own hands and legalizing retaliations leads to a risky situation for anyone caught in the middle. With so many variables, the next section explores unbiased arguments for both positions.

Stakeholder Argument 1: The Cyber Attack Victim

Being the victim of a cyber-attack can very stressful. With a cyber-attack, comes the feeling of insecurity, especially if something as sensitive as data is stolen. To rectify this invasion of privacy, the instinct is to take back what is yours. From the perspective of the hacked victim, the ethical framework that best resonates with their reasoning is the Virtue. Virtue ethics revolves around being morally justified. The ideas of the Ethical framework of Virtue are rooted in the teachings of Plato and Aristotle. The main idea revolves around a sense of character and prioritizing virtues such as justice, wisdom, and courage. Under the ethical virtue framework, consequences are overlooked in favor of developing virtues. In this case, the sense of justice is highlighted. By highlighting the virtue of justice, we can see why seeking attribution becomes paramount.

For the victim, the biggest upside of retaliation is minimizing the loss of data. In cyberspace, time is valuable. For each minute, the data is not under control it could be on auction in a matter of minutes. It is in the best interest of the victim to seek attribution, neutralize the attacker, and minimize the possibility of data breach or leak. (Nachreiner, 2024).

Stakeholder Argument 2: The Argument of the Innocent

Imagine an ordinary day and you wake up to your device being corrupted, systems compromised and essentially have a bricked device/ system. This same scenario is even more devastating to a business. An inoperable machine could mean system downtime and lost profits for the day. The unintentional targeting and destruction of property are some of the worst fears of the second stakeholder, the innocent bystander getting caught in the crossfire between the hacker and the hacker. In an act of retaliation, a log or digital paper trail of connections could pinpoint

an innocent party. With so much harm that could be done, their stance is against the legalization of hacking back. According to an article from SC Media, "…. The main risks (although certainly not the only ones) are that: (1) hacking back can quickly cross the fine line separating legal vs. illegal activities, and (2) that you risk a high chance of inflicting collateral damage, mostly since attacker attribution is highly inaccurate in many cases." (Barak, 2017).

Defending the peace and appealing to the greater good revolves around the Utilitarianism ethical framework. Utilitarianism was initially coined by Jeremy Bentham and was built upon by philosophers such as John Stuart Mill, and Henry Sedwick. The main idea circles around the greater good, regardless of morality. When applied to something such as cybersecurity, we must examine who is at risk and the potential damage that could be done if hacking back were to be legalized. Under the influence of Utilitarianism, it isn't a matter of what is wrong or right, but rather what keeps the peace and keeps collateral damage to a minimum. One can argue that legalizing hacking back with the intent to retaliate would encourage cyber vigilantism. By allowing hacking back, it could possibly induce a Hammurabi law-esque digital landscape: an eye for an eye (or for a more relevant comparison: a corrupt database for a corrupt database). Despite that being a slippery slope type of claim, the increased possibility of innocent parties getting caught in the crossfire is a big problem. Laws and protocols are in place for security breaches and the proper authorities should be alerted. To draw parallels to the physical world, the authorities are provided training and given specialized tools to handle the situation. Giving the same freedom of retaliation in the real world could lead to many people getting hurt; especially if one is not trained to handle a hostile situation.

Personal Opinion

Hacking back should be legal (to an extent). When excessive force is applied, the lines between offensive and defensive maneuvers become blurred. Prior to doing all this research, I would have sided with legalizing retaliation. However, after reading about the potential risks and harm that could be inflicted on innocent users, my views on the subject have changed. Although my opinion leans toward agreeing with the victim, I believe that there is a more responsible approach. Being able to defend oneself is a right. No one should lay on the ground belly up and take abuse regardless of if it is in the physical world or cyberspace.

My proposal requires the rest of the world to be on the same page in terms of cybersecurity. Currently, hacking back is illegal and the proper way to report a cybercrime attack is to first alert the local FBI cyber team, Cybersecurity and Infrastructure Security Agency. Per their documentation "A private sector entity that is a victim of a cyber incident can receive assistance from government agencies, which are prepared to investigate the incident, mitigate its consequences, and help prevent future incidents." (CISA). But as mentioned earlier, time is of the essence, and stolen data can be on sale on the black market in a matter of minutes. With something as time sensitive as this, measures can be taken defensively to minimize and mitigate a data breach. Similar to how there are fire safety drills, earthquake drills, or school shooter drills, creating a standard exit plan in case of a disaster can minimize the damage done during an attack. Actively monitoring networks for malicious activity, setting up system isolation mechanisms, and quicker incident responses are all defensive measures that can be taken to protect oneself from further harm. If the unauthorized user is still connected to the victim's digital device. I believe that hacking back is justified only for attribution and identification of the machine that infiltrated the attack. Any further harm enters this grey area, and it becomes difficult to distinguish if the hack-back is only for attribution of data or if sabotage of the other

parties' devices was involved. The moment where a hack-back starts destroying another party's property, then it no longer becomes a justified form of self-defense. Rather it in itself becomes a cyber-attack. These fine lines are what make differentiating between self-defense and attack. This is why a legal rework is necessary: to properly define these lines.

Additionally, I believe it is important to have a unified global legal framework against cyberattacks. Specifics such as what servers were accessed and hijacking of third parties change how the law affects the victim, and all other parties involved. Just as technology advances rapidly, the legal system should be able to accommodate and clearly define cybercrimes. Without clear legal definitions, it would lead to an endless blame cycle of pointing fingers back at one another; all while the actual attacker gets away without a scratch. Hackers do not abide by national state or federal borders. And this is why there needs to be a global legal framework for cybersecurity.

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